ModelIt!

Teacher's Guide

Why Hot Cheetos Make You Cry

How Your Body's Alarm System Detects and Responds to the World

Quick Reference

Grade Level	Grade 7
NGSS Standard	MS-LS1-8
Time	45–60 minutes (core lesson)
Materials	Student Activity Pack, devices with ModelIt! access
Prep	Print activity packs, test ModelIt! on student devices
Grouping	Partners or small groups (2–4)

ModelIt! — Discovery Collective

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Table of Contents

ModelIt!

1 NGSS Standards Unpacking & CAST Alignment 3

2 Scoring Rubric 4

3 Background Content Knowledge 5

4 The LEVER Framework 6

5 Slide-by-Slide Facilitation Guide 7

6 Answer Key & Expected Responses 9

7 Collaborative Learning Protocols 10

8 Differentiation & Extensions 11

9 Common Misconceptions 12

1 NGSS Standards & CAST Alignment

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Performance Expectation: MS-LS1-8

Gather and synthesize information that sensory receptors respond to stimuli by sending messages to the brain for immediate behavior or storage as memories.

Three-Dimensional Learning

Science Practice	**Obtaining, Evaluating, and Communicating Information** — Students gather and synthesize information about how sensory receptors detect stimuli and communicate with the brain.
Disciplinary Core Idea	**LS1.D Information Processing** — Each sense receptor responds to different inputs (light, sound, chemicals), transmitting them as signals that travel along nerve cells to the brain for processing.
Crosscutting Concept	**Cause and Effect** — Students identify that specific stimuli CAUSE specific receptor activations, which trigger predictable nervous system responses.

CAST Testing Alignment

MULTIPLE CHOICE:

A person eats a Hot Cheeto and immediately starts sweating and tearing up. This response occurs because capsaicin in the chip:

CONSTRUCTED RESPONSE:

Describe the pathway a signal takes from the moment you touch something hot to the moment you pull your hand away. Explain the role of sensory receptors, neurons, and the brain in this process.

2 Scoring Rubric

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Assessment Reference

This is the same rubric students see in their Activity Pack. Use it during and after the lesson for formative and summative assessment. Circle or highlight the proficiency level for each skill.

Skill	Expert (4)	Proficient (3)	Developing (2)	Beginning (1)
Identifying Components	EXPERT Correctly identifies and sorts ALL components as internal or external with detailed reasoning	PROFICIENT Correctly identifies and sorts MOST components with some reasoning	DEVELOPING Identifies components but has difficulty sorting or explaining	BEGINNING Needs significant help identifying components
Establishing Relationships	EXPERT Accurately describes ALL positive and negative relationships with clear evidence	PROFICIENT Accurately describes MOST relationships with some evidence	DEVELOPING Identifies some relationships but struggles with direction	BEGINNING Needs help understanding how components connect
Running Simulations	EXPERT Makes accurate predictions AND explains surprising results using evidence	PROFICIENT Makes reasonable predictions and records clear observations	DEVELOPING Runs simulations but predictions or observations are incomplete	BEGINNING Needs help running simulations or recording data
Explaining Results	EXPERT Uses multiple pieces of evidence to explain HOW and WHY systems interact	PROFICIENT Uses evidence to explain system interactions	DEVELOPING Gives partial explanations with limited evidence	BEGINNING Needs help connecting evidence to explanations

Scoring Guidance

Expert (4): Student demonstrates deep understanding and can transfer knowledge to new contexts.
Proficient (3): Student meets grade-level expectations and shows solid understanding.
Developing (2): Student is making progress but needs additional scaffolding or practice.
Beginning (1): Student needs significant support; consider using Differentiation strategies (Section 7).

3 Background Content Knowledge

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For Your Confidence

This section provides the science background you need to facilitate this lesson with authority. You don't need to teach all of this — it's here so you can answer student questions accurately.

Your nervous system is the most sophisticated information processing system ever evolved. It detects millions of stimuli per second, transmits signals at highway speeds, and coordinates responses that keep you alive — all without you having to think about it.

Sensory Receptors

The human body contains dozens of receptor types: photoreceptors (light), mechanoreceptors (pressure/vibration), thermoreceptors (temperature), chemoreceptors (taste/smell), and nociceptors (pain). Each type responds to a specific stimulus and converts it into electrical nerve signals.

The Capsaicin Trick

Capsaicin, the compound that makes peppers hot, binds to TRPV1 receptors — the same receptors that detect temperatures above 43°C (109°F). When capsaicin activates these receptors, the brain receives a signal indistinguishable from actual burning. The response is real (tears, sweating, pain) even though no tissue damage occurs.

Signal Transmission

Nerve signals are electrical impulses that travel along neurons at speeds of 1-120 m/s depending on the nerve type. The signal passes from neuron to neuron across tiny gaps called synapses using chemical neurotransmitters. From stubbed toe to brain awareness takes about 0.1 seconds.

Brain Processing

The brain receives sensory input and decides the response: ignore it (background noise), store it (memories), or act on it (pull hand away). The somatosensory cortex maps which body part sent the signal. Pain signals get priority processing — your brain is wired to respond to danger first.

4 The LEVER Framework

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About LEVER

LEVER is how every ModelIt! lesson works. Students move through five phases to build, test, and refine a computational model. Each lesson is self-contained — students complete all five phases in one session.

L **Locate the System**
Students identify stimulus intensity, receptor sensitivity, nerve signal strength, and brain response as components of the sensory system.

E **Establish Relationships**
Students determine that stimulus intensity and receptor sensitivity both positively affect nerve signal strength, which positively affects brain response.

V **Visualize & Model**
Students build a model showing how environmental stimuli are converted to nerve signals and brain responses.

E **Evaluate Outcomes**
Students run mild, capsaicin, and adaptation scenarios to observe how the system responds to different inputs.

R **Revise & Extend**
Students add reflex arc speed or sensory adaptation to explore more complex nervous system behaviors.

Dropping ModelIt! Into Your Existing Unit

ModelIt! is a **supplement**, not a replacement for your science curriculum. Each lesson stands alone, but it becomes even more powerful when students revisit their model throughout a unit. Here are easy ways to weave it in:

At the start of a unit	Use this lesson to introduce the phenomenon. Students build a “first draft” model based on what they already know. Save it.
After a reading or lab	Have students reopen their model and update it: “Based on what you just learned, would you add a component? Change a relationship?” Takes 10–15 minutes.
As a “what-if” tool	When students ask questions during class, say: “Great question — go test that in your model!” Students run new scenarios and report back.
For peer discussion	Partners compare models: “Why does your model have this arrow and mine doesn’t?” Differences spark scientific debate using evidence.
As a presentation tool	Students project their model and walk the class through it live — showing components, running scenarios, answering audience “what-if” challenges in real time.
At the end of a unit	Compare Day 1 model to final model. The growth IS the assessment — students see how their thinking evolved.

5 Slide-by-Slide Facilitation

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Slide 1: Cover

2 min

Say: *"Why do Hot Cheetos make you cry, sweat, and feel like your mouth is literally on fire?"*

Watch For

Show a video of someone trying extremely spicy food. The reaction is real — but the 'fire' isn't.

Slide 2: Learning Objectives

3 min

Say: *"Today we're investigating how your body's alarm system works — and how it can be tricked."*

Watch For

Have students read objectives. Pre-teach 'stimulus' and 'receptor.'

Slide 3: Big Question

3 min

Say: *"Right now your body is detecting thousands of things — temperature, light, sound, pressure. How?"*

Watch For

Quick activity: close eyes. What can you still detect? Touch, sound, temperature.

Slide 4: LEVER Framework

2 min

Say: *"We'll model the path from stimulus to response."*

Watch For

Preview LEVER steps.

Slide 5: Activity 1: Components

8 min

Say: *"What starts a nerve signal? What controls how strongly you react?"*

Watch For

Guide sorting. Discuss why stimulus and receptor sensitivity are inputs.

Slide 6: Activity 2: Connections

8 min

Say: *"Stronger stimulus, stronger signal, bigger response. But is it always that simple?"*

Watch For

Help students build the signal pathway chain.

Slide 7: Activity 3: Simulation

10 min

Say: *"Let's simulate what happens when capsaicin hits your tongue at full force!"*

Watch For

Students predict first. Compare mild vs. capsaicin scenarios.

Slide 8: Discoveries

5 min

Say: *"So capsaicin doesn't actually burn you — it TRICKS your pain receptors. What else does this?"*

Watch For

Discuss sensory tricks. Mention menthol (cold receptor trick).

Slide 9: STEM Challenge

12 min

Say: *"You're neuroscientists now. Design an experiment to test sensory perception."*

Watch For

Groups design and conduct sensory experiments.

6 Answer Key & Expected Responses

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Component Sorting

- **EXTERNAL:** Stimulus Intensity, Receptor Sensitivity
- **INTERNAL:** Nerve Signal Strength, Brain Response

Relationships

- **Stimulus Intensity to Nerve Signal Strength:** POSITIVE (+) — A stronger stimulus activates more receptors and triggers higher-frequency nerve impulses, creating a stronger signal.
- **Receptor Sensitivity to Nerve Signal Strength:** POSITIVE (+) — More sensitive receptors fire at lower thresholds and generate stronger signals for the same stimulus level.
- **Nerve Signal Strength to Brain Response:** POSITIVE (+) — Stronger nerve signals cause more intense brain responses — from barely noticing to full fight-or-flight reactions like crying and sweating.

Key Discoveries

- Different sensory receptors are specialized for different stimuli — you have separate detectors for heat, cold, pressure, and chemicals
- Capsaicin doesn't actually burn you — it activates the SAME heat receptors that real fire would, tricking your brain
- Nerve signals travel at up to 120 meters per second — that's faster than a car on the highway
- Your brain processes millions of sensory inputs simultaneously and decides which ones deserve a response

7 Collaborative Learning Protocols

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Why This Matters

ModelIt! is a digital tool, but the deepest learning happens between humans. Computational modeling is most powerful when students build understanding *together* — debating relationships, challenging predictions, and constructing explanations as a community. These protocols ensure every student's voice is part of the model. Use at least ONE per lesson.

Model Gallery Walk

When: After students build models (LEVER Phase V)
Setup: Groups post their models on screens/desks. Half the class stays to "defend," half rotates with sticky notes.
Protocol: Visitors leave 2 sticky notes: one "I notice..." and one "I wonder..." After rotation, defenders read feedback aloud. Class discusses patterns.
Why it works: Students see that the SAME phenomenon can be modeled differently — and that's how real science works. (Bandura: learning through observation of peers)

Scientist Fishbowl

When: During simulation discussion (LEVER Phase E)
Setup: 4-5 students sit in center circle. Rest of class observes with clipboards.
Protocol: Inner circle discusses: "What happened in Scenario 3 and WHY?" using only evidence from their models. Outer circle records strongest arguments. Swap groups.
Why it works: Students practice scientific discourse without teacher directing. Observers learn to evaluate arguments. (Freire: dialogue as the foundation of critical consciousness)

Component Jigsaw

When: During component analysis (LEVER Phase L)
Setup: Assign each group ONE component to become "experts" on. They research it, discuss it, draw it.
Protocol: Regroup so each new team has one expert per component. Experts teach their component to the team. Team builds the model together using everyone's expertise.
Why it works: Every student is essential — the model literally cannot be built without each person's knowledge. Interdependence drives engagement. (Lave & Wenger: learning through legitimate peripheral participation)

Prediction Debate

When: Before running simulations (LEVER Phase E)
Setup: Post a scenario. Students physically move to corners: "Fire Spread will INCREASE a lot" / "INCREASE a little" / "STAY the same" / "DECREASE."
Protocol: Each corner group has 2 minutes to build their argument using model logic. One spokesperson presents. Then RUN the simulation. Losing corners explain what they missed.
Why it works: Movement + social risk + immediate feedback. Students CARE about the outcome because they committed publicly. Wrong predictions become the best learning moments.

Pair Programming for Models (Digital Collaboration)

When: Throughout model building
Setup: Two students, one device. **Driver** controls the mouse/keyboard. **Navigator** directs strategy and catches errors. Switch every 5 minutes.
Protocol: Navigator must verbalize every decision: "I think we should add a negative arrow from Rainfall to Dry Vegetation because..." Driver cannot act without Navigator's reasoning.
Why it works: Forces articulation of scientific thinking. The model becomes a shared artifact, not an individual assignment. Students who struggle with writing can demonstrate understanding verbally. Borrowed from computer science — where the best code is written in pairs.

Quick Protocol Selection Guide

Short on time (5 min)	Prediction Debate — gets bodies moving, generates energy fast
Full period available	Component Jigsaw — deep collaboration, every student contributes

After simulation day	Model Gallery Walk — peer feedback, multiple perspectives
Advanced discussion	Scientist Fishbowl — builds academic discourse skills
Every day default	Pair Programming — transforms individual work into dialogue

8 Differentiation & Extensions

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Below Grade Level

- Pre-sort components as a whole class
- Provide sentence frames for relationships
- Use visual vocabulary cards
- Pair with a strong reader

Advanced Learners

- Add 2-3 additional components
- Research real-world data
- Create multi-step causal chains
- Compare to similar systems globally

English Language Learners

- Pre-teach key vocabulary
- Provide bilingual glossary
- Use visual supports with text
- Allow verbal responses

Gifted & Talented

- Create feedback loop diagrams
- Research prevention strategies
- Write a policy brief
- Connect to climate change

9 Common Misconceptions

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Why This Matters

Addressing misconceptions proactively helps students build accurate scientific understanding. Use these as formative assessment checkpoints during the lesson.

STUDENTS MAY THINK	SCIENTIFIC UNDERSTANDING
"Spicy food burns your mouth"	Capsaicin doesn't cause any tissue damage or burning. It binds to heat receptors (TRPV1) and tricks your brain into THINKING you're being burned. The pain is real, but the injury is not.
STUDENTS MAY THINK	**SCIENTIFIC UNDERSTANDING**
"We only have five senses"	Humans have far more than five senses. Beyond sight, hearing, taste, smell, and touch, we have proprioception (body position), vestibular sense (balance), thermoception (temperature), nociception (pain), and more.
STUDENTS MAY THINK	**SCIENTIFIC UNDERSTANDING**
"Pain is always bad"	Pain is actually a critical survival system. It alerts your brain to danger and triggers protective responses. People born without pain receptors (a real condition) suffer constant injuries because they can't detect damage.